

HiveForce Labs

# THREAT ADVISORY



## ATTACK REPORT

**Reaper, North Korean hacking group, targets defectors**

Date of Publication

March 20, 2023

Admiralty Code

A1

TA Number

TA2023145

# Summary

**First Seen:** 2012

**Threat Actor:** Reaper (APT 37, Ricochet Chollima, ScarCruft, Thallium, Group 123, Red Eyes, Geumseong121, Venus 121, Hermit, InkySquid, ATK 4, ITG10)

**Attack Countries:** China, Czech, Hong Kong, India, Japan, Kuwait, Nepal, Poland, Romania, Russia, South Korea, UK, USA, Vietnam

**Attack Industry:** Aerospace, Automotive, Chemical, Financial, Government, Healthcare, High-Tech, Manufacturing, Technology, Transportation, Defectors, NGOs

**Malware:** Chinotto

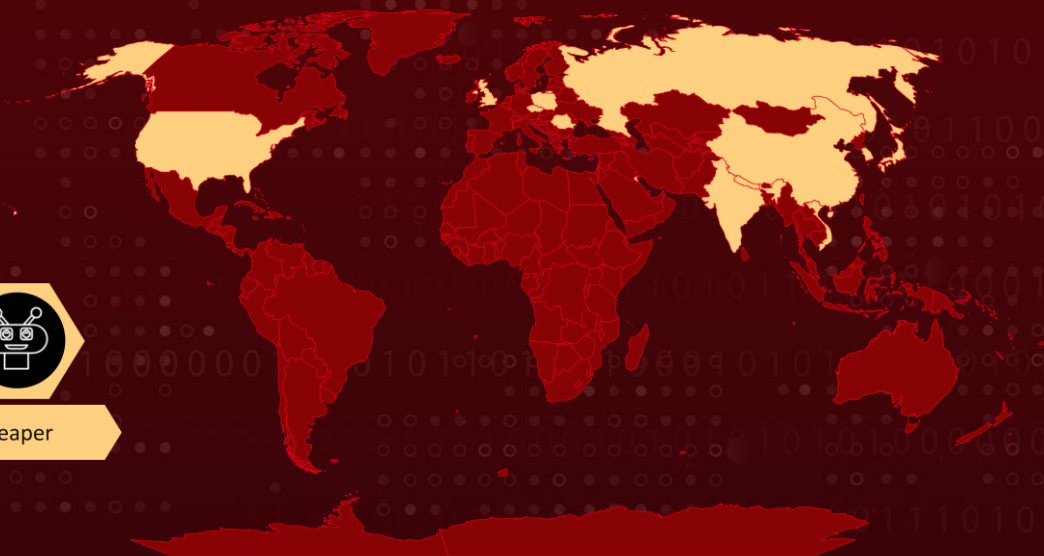
**Attack:** North Korea-linked hacking group Reaper has been actively conducting cyber espionage campaigns against NGOs and civil society since at least 2012, and they have recently been observed engaging in surveillance of North Korean defectors.



## Attack Regions



Reaper



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# Attack Details

## #1

According to the investigation, two open directories were discovered on the Command-and-Control servers used by the North Korean-linked hacking group Reaper, also known as APT37. These directories allowed the investigators to view the malicious software that was hosted on the servers as well as the data that had been stolen from the group's victims. The group has been active since at least 2012, primarily conducting cyber espionage campaigns against NGOs and civil society.

## #2

The investigation uncovered several interesting findings, including several phishing webpages, a CHM infection vector, new PowerShell implants, and Chinotto malware modules. Chinotto malware is a type of malware that was first documented in November 2021 and has since been associated with the Reaper group. It is noteworthy that the investigation uncovered new PowerShell implants, which are powerful tools used by attackers to carry out malicious activities.

## #3

Based on the observed activity, it is highly likely that the group is currently conducting surveillance on North Korean defectors. This activity is associated with high confidence to Reaper, and it is possible that the group is using their cyber espionage capabilities to identify and target defectors.

# Recommendations



### Security Leaders

Phishing simulations and routine education and awareness training and communications rarely account for MFA fatigue and web browser hygiene. Integrate and communicate all lessons learned.



### Security Engineers

- **Uni5 Users:** This is an actionable threat advisory in HivePro Uni5. Prioritize and block all indicators attributed to the threat actors and attacks through your Command Center. Test your controls with Uni5's Breach & Attack Simulation.
- **All Engineers:** Refer to and action upon the 'Potential MITRE ATT&CK TTPs' & 'Indicators of Compromise (IoC)' on the following pages.

# Potential MITRE ATT&CK TTPs

|                                                                |                                                              |                                                          |                                                               |
|----------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------|
| <b><u>TA0042</u></b><br>Resource Development                   | <b><u>TA0001</u></b><br>Initial Access                       | <b><u>TA0002</u></b><br>Execution                        | <b><u>TA0003</u></b><br>Persistence                           |
| <b><u>TA0005</u></b><br>Defense Evasion                        | <b><u>TA0006</u></b><br>Credential Access                    | <b><u>TA0007</u></b><br>Discovery                        | <b><u>TA0009</u></b><br>Collection                            |
| <b><u>TA0010</u></b><br>Exfiltration                           | <b><u>TA0011</u></b><br>Command and Control                  | <b><u>T1566</u></b><br>Phishing                          | <b><u>T1566.001</u></b><br>Spearphishing Attachment           |
| <b><u>T1204</u></b><br>User Execution                          | <b><u>T1212</u></b><br>Exploitation for Credential Access    | <b><u>T1555</u></b><br>Credentials from Password Stores  | <b><u>T1555.003</u></b><br>Credentials from Web Browsers      |
| <b><u>T1036</u></b><br>Masquerading                            | <b><u>T1036.005</u></b><br>Match Legitimate Name or Location | <b><u>T1059</u></b><br>Command and Scripting Interpreter | <b><u>T1059.001</u></b><br>PowerShell                         |
| <b><u>T1059.005</u></b><br>Visual Basic                        | <b><u>T1056</u></b><br>Input Capture                         | <b><u>T1056.001</u></b><br>Keylogging                    | <b><u>T1113</u></b><br>Screen Capture                         |
| <b><u>T1584</u></b><br>Compromise Infrastructure               | <b><u>T1584.006</u></b><br>Web Services                      | <b><u>T1547</u></b><br>Boot or Logon Autostart Execution | <b><u>T1547.001</u></b><br>Registry Run Keys / Startup Folder |
| <b><u>T1140</u></b><br>Deobfuscate/Decode Files or Information | <b><u>T1033</u></b><br>System Owner/User Discovery           | <b><u>T1082</u></b><br>System Information Discovery      | <b><u>T1560</u></b><br>Archive Collected Data                 |
| <b><u>T1560.002</u></b><br>Archive via Library                 | <b><u>T1071</u></b><br>Application Layer Protocol            | <b><u>T1071.001</u></b><br>Web Protocols                 | <b><u>T1573</u></b><br>Encrypted Channel                      |
| <b><u>T1573.001</u></b><br>Symmetric Cryptography              | <b><u>T1041</u></b><br>Exfiltration Over C2 Channel          |                                                          |                                                               |

## ✂ Indicators of Compromise (IOCs)

| TYPE   | VALUE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URLs   | <p>hxxp[:]//141.105.65[.]165/data/<br/>hxxp[:]//141.105.65[.]165/files/<br/>hxxp[:]//141.105.65[.]165/main/<br/>hxxp[:]//141.105.65[.]165/support/<br/>hxxp[:]//attiferstudio[.]com/install.bak/sony/<br/>hxxp[:]//ri-guard[.]com/download/temp/cn-var/<br/>hxxp[:]//koaagj.co[.]kr/files/2014/12/fix/<br/>hxxp[:]//jdwanxiang[.]com/win/shenti/<br/>hxxps[:]//clovery-shapes.000webhostapp[.]com/defcon/<br/>hxxp[:]//hk-law.co[.]kr/data/file/joomla/<br/>hxxp[:]//172.93.193[.]158/data/</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| SHA256 | <p>e96a18b5837c7a7d83215d70ca10b84ee8c7b6e8dbd4d215586ec062d328ce86,<br/>1304fbeca197e4e67959c0b89b619cc109e4825d0da26ac41277eb34d2a19bc6,<br/>1409c4d0bd9a22a1e5adf016fffb83bbf3bd9f72ae0773780a409b980ed97763,<br/>6c1f0deadbfe5aede933592a9692b18879232a29bfdda5a666b91475b4746612,<br/>c9ea7afef5ac790297bdd0fb78c06186516957542e5da326075a0e2d230c27c1,<br/>a320ef003f43b28960043f95076c2066891e3a6a785476a2615a1f7b50a11c78,<br/>a88dc9a152cc7758a1df5aa33cf7b31cdb14e593a8744f2059602a49b8b04e0f,<br/>1fcf8bfcd70b97c6d3c9ac93602db4ee41a5d09f0a4b92fa67b76668fb33811d,<br/>309cb38fbcd0132552fc739dd37d32c24b91ba712bebb9886d4638bceb2d8bf3,<br/>001e8e66fa4afc58cab23f5ee490f3080ef985c83d3b2a4555fc9c39cfe56bd8,<br/>3f0f0060bebd891008eaf8a647d91803107fc52294ceaab8b59b89958db4a0de,<br/>58cdb73495c2d6120f81f3752828f532b6a70ce7617b725e6989cf2d9cdb6aa4,<br/>5e67b5aed329e6545a36eff46bb6db8bdaa17841a4ec77228955d81872fed549,<br/>ccb6b0e02f7a9d7a541d0ed352706de943c178650a675eff667cb848ec0dc977,<br/>c0a36e340cc38c9abd07029e3d621395575c9a4a64459334ef84b623d1058865,</p> |

| TYPE   | VALUE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SHA256 | 40341da349e684593bbd01b244f94c28aa024ee49c3b3ebc89960e53e40750ae,<br>9c30265e5b8f7b6017141815001a0678a99d05dc8302b50517c47d5f282b3c36,<br>575631e9548b0f91addf3ce68bc5b4b9e86a17c069a221815062e1aa93d2978e,<br>cd2028bf873293aa330eb21ab96f8e71fa91e2d212852e81f292e769c0fba2d7,<br>2a22da882f05dc159b055b01de7331e22fb6b5ce858308015a912b49487c56f0,<br>b4a8d58b5d5e49d9d65b8b9faba344923cbac87473f2a32c646e359799ed655e,<br>3d6c99e137bf5653134049caf9010d4c6d82360bf569ecca05fe15440d7fd0b0,<br>ff5fc46c9598a5fbc9b54fbbd05ee0bb86549ecbba715093413326b58a1b9d2b,<br>b364bac52981edd74fbc45cca4216e66da5df9918000cc4617156ab42c914e7e,<br>b73ee977154402f8eccc5a446baf0dba456a37d1ca9348858540a8d048f3fd37,<br>2fa36a4eb676f3afb1774224bc59041944ddfa4a3417630d01659ba3f0ced834,<br>ead97a3920ff557299bcd4ccde1770c759263b93b70414258ec9030bbd0cb750,<br>bdb33062bddd53043bab508e8e96b7c8353549d8eaa4b9004e7b3303e8a4e91b,<br>0b6202a043f8dcf0690660c5c8f7a75f07336ca5576164295da34a569129548f,<br>8078fc582b8f5eb81ef5d8da2b11fb4ce63f52fdeb1c2ceb3ac7a01113f9f3ee,<br>60804ebbb655ea68b9e0bce63d5edbd03e0f75837f44539fec28dc12d44b5ba5,<br>e6d9c5a401a733ceb80b004deb347092affe572eda4e1ca6aa6c77bb0c6ea7e8 |



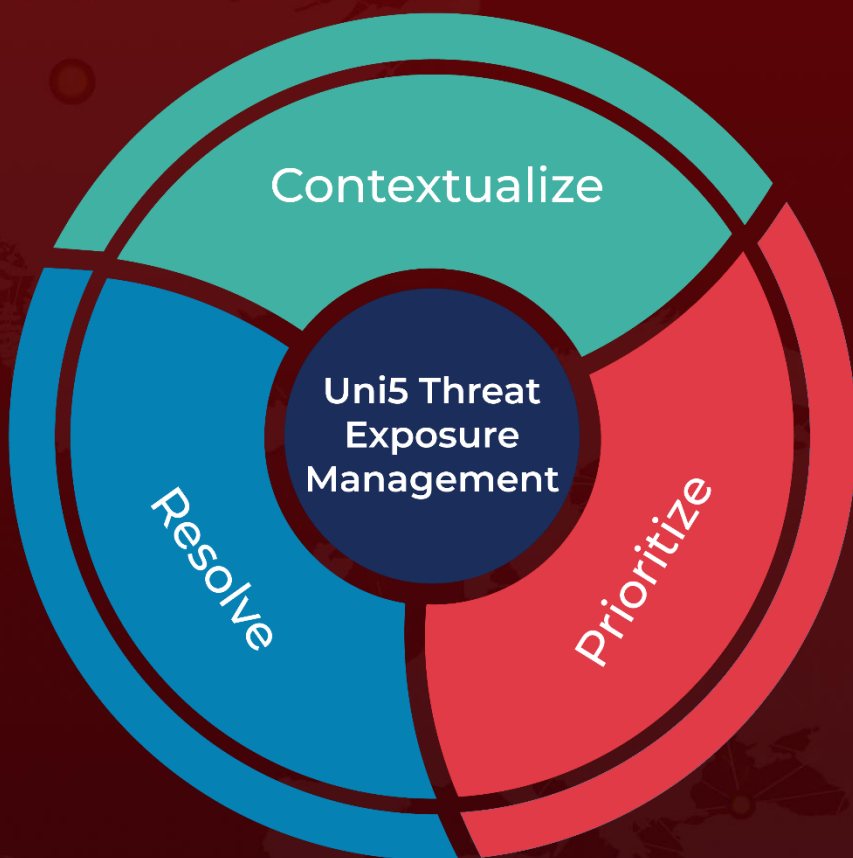
## References

<https://blog.sekoia.io/peeking-at-reaper-surveillance-operations-against-north-korea-defectors/>

# What Next?

At Hive Pro, it is our mission to detect the most likely threats to your organization and to help you prevent them from happening.

Book a free demo with HivePro Uni5: Threat Exposure Management Platform.



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**March 20, 2023 • 5:00 AM**

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